

OBJECTIVE

To obtain a research career in Computer Science and Engineering possessing effective and dynamic research skills along with proficient technical skills.

EDUCATION

The University of Arizona (UofA) Ph.D. in Computer Science, CGPA: 3.889/4.00 – Research Focus: Natural Language Processing – Advisor: Dr. Mihai Surdeanu, Professor	Arizona, USA Fall 2021–Present
Bangladesh University of Engineering and Technology (BUET) M.Sc. in Computer Science and Engineering, CGPA: 3.75/4.00 – Thesis: “Devising A Selective Sensor Deployment Strategy Leveraging NSGA-II for Detecting Derailments in the Railway System of Bangladesh” – Advisor: Dr. A. B. M. Alim Al Islam, Professor	Dhaka, Bangladesh November 2018–August 2021
Bangladesh University of Engineering and Technology (BUET) B.Sc. in Computer Science and Engineering, CGPA: 3.58/4.00 – Thesis: “Enhancing Fidelity of Quantum Cryptography using Maximally Entangled Qubits” – Advisor: Dr. A. B. M. Alim Al Islam, Professor	Dhaka, Bangladesh July 2014–October 2018

PROFESSIONAL EXPERIENCE

The University of Arizona Research Assistant, Department of Computer Science (Fall 2022, Fall 2023)	Arizona, USA Fall 2021 –Present
The University of Arizona Teaching Assistant, Department of Computer Science – CSC 380: Principles of Data Science (Fall 2021, Spring 2022, Spring 2023, Fall 2023, Spring 2026) – CSC 372: Comparative Programming Languages (Spring 2024, Fall 2024) – CSC 473: Automata, Grammars and Languages (Spring 2025) – CSC 337: Web Programming (Fall 2025)	Arizona, USA
Bangladesh University of Engineering and Technology Research Assistant at NeC Lab, Department of Computer Science and Engineering – PI: Dr. A. B. M. Alim Al Islam, Professor	Dhaka, Bangladesh November 2018–July 2021

RESEARCH INTEREST

My research broadly focuses on ML and Security, covering the following:

- Natural Language Processing
- Machine Learning
- Computer Networks and Security
- Optimization

PROFESSIONAL SERVICES

- Reviewer: USENIX Artifact Evaluation '22, '23; EMNLP '24; NeurIPs '25;
- Sub Reviewer: NDSS '22, '23; PoPETs '22; USENIX '23; IEEE S&P '24; ICLR '26;

AWARDS AND RECOGNITIONS (SELECTIVE)

- **Graduate College Fellowship** (The University of Arizona) Fall 2021, Spring 2023, Spring 2024, Spring 2025, Spring 2026
- **ICT Fellowship** (ICT Division, Ministry of Posts, Telecommunications and Information Tech., Bangladesh) 2019 - 2021
- **BUET Technical Scholarship** 2014 - 2018
- **5th Position** (Nationwide AI-Based Dhaka Traffic Detection Challenge) 2020
- **1st Runner-up** (Hand written digit recognition competition, National Robotech Festival) 2017

EXTRACURRICULAR ACTIVITIES (SELECTIVE)

• Volunteering EMNLP	2024
• Volunteering ACM ICPC Rocky Mountain Regional Qualifier	2024
• Member of InfoSarker Survey Team (Bangladesh National Internet Survey)	2019 - 2020
• Lead Organizer of Inter University Project Show Competition of BUET CSE Festival	2018
• Founder and Lecturer at Mathematics and ICT Coaching Center	2016 - 2018

PUBLICATIONS (SELECTED)

CONFERENCE

1. **Saiful Islam Salim**, Enrique Noriega Atala, Haris Riaz, Marvin J. Slepian, and Mihai Surdeanu, “*PathwayLM: Multi-hop Mechanistic Pathways for Biomedical Language Model Reasoning*”, Association for Computational Linguistics (ACL). (Under Review)
2. **Saiful Islam Salim**, Rubin Yuchan Yang, Alexander E Cooper, Saumya Debray, and Sazzadur Rahaman, “*Impeding LLM-assisted Cheating in Introductory Programming Assignments via Adversarial Perturbation*”, In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP), Miami, Florida, USA. Association for Computational Linguistics. (Oral Presentation; Nominated for the Best Paper)
3. Masfiquir Rahaman, MD. Nazmul Hasan Sakib, Nafisa Islam, **Saiful Islam Salim**, Uday Kamal, Raihan Rasheed, and A. B. M. Alim Al Islam, “*Let’s Vibrate with Vibration: Augmenting Structural Engineering with Low-Cost Vibration Sensing*”, EAI MobiQuitous ’23, Melbourne, Australia, 2023.
4. **Saiful Islam Salim**, Uday Kamal, Adnan Quaium, Mainul Hossain, Masfiquir Rahaman, Nazmul Hasan Sakib, and A. B. M. Alim Al Islam, “*Long-Range Low-Cost Networking for Real-Time Monitoring of Rail Tracks in Developing Countries*”, ACM ICTD ’22, Washington, USA, 2022.
5. Tarik Reza Toha, Masfiquir Rahaman, **Saiful Islam Salim**, Mainul Hossain, A. M. Sadri, and A. B. M. Alim Al Islam, “*DhakaNet: Unstructured Vehicle Detection using Limited Computational Resources*”, IEEE ICDM ’21, Auckland, New Zealand, 2021.
6. Ashish Sutradhar, Md Harunur Rashid Moon, Faria Tasneem, Pritom Goswami, Tarik Reza Toha, **Saiful Islam Salim** and Shaikh Md. Mominul Alam, “*Devising a Dust and Noise Pollution Monitoring System for Textile Industry*”, ACM NSysS ’21, Dhaka, Bangladesh, 2021.
7. **Saiful Islam Salim**, Adnan Quaium, Sriram Chellappan, and A. B. M. Alim Al Islam, “*Enhancing Fidelity of Quantum Cryptography using Maximally Entangled Qubits*”, IEEE GLOBECOM ’20, Taipei, Taiwan, 2020.
8. Sadia Sharmin, **Saiful Islam Salim**, and Kazi Ragib Ishraq Sanim, “*A Low-cost Urban Search and Rescue Robot for Developing Countries*”, IEEE RAAICON ’19, Dhaka, 2019.

JOURNAL

1. **Saiful Islam Salim**, Adnan Quaium, Uday Kamal, Masfiquir Rahaman, Mainul Hossain, Nazmul Hasan Sakib, Tarik Reza Toha, and A. B. M. Alim Al Islam, “*A cost-effective real-time rail track monitoring system leveraging multi-sensor fusion and multi-objective optimization*”, Scientific Reports, Nature, 2025. (Q1 in Scimago; IF: 3.9)
2. Md. Masum Mushfiq, Tarik Reza Toha, **Saiful Islam Salim**, Aaiyeesha Mostak, Masfiquir Rahaman, Najla Abdulrahman Al-Nabhan, Arif Mohamin Sadri, and A. B. M. Alim Al Islam, “*To Lane or not to Lane? Comparing on-road Experiences in Developing and Developed Countries using a new Simulator ‘RoadBird’*”, IEEE Transactions on Intelligent Transportation Systems, IEEE, 2024. (Q1 in Scimago; IF: 7.9)
3. Adnan Quaium*, Najla Abdulrahman Al-Nabhan*, Masfiquir Rahaman*, **Saiful Islam Salim***, Tarik Reza Toha, Jannatun Noor, Mainul Hossain, Nafisa Islam, Aaiyeesha Mostak, Md Shihabul Islam, Md. Masum Mushfiq, and A. B. M. Alim Al Islam, “*Towards Associating Negative Experiences and Recommendations Reported by Hajj Pilgrims in a Mass-Scale Survey*”, Heliyon, Elsevier, 2023. (Q1 in Scimago; IF: 4.0)
4. Tarik Reza Toha, Najla Abdulrahman Al-Nabhan, **Saiful Islam Salim**, Masfiquir Rahaman, Uday Kamal, A. B. M. Alim Al Islam, “*LC-Net: Localized Counting Network for Extremely Dense Crowds*”, Applied Soft Computing (ASOC), Elsevier, 2022. (Q1 in Scimago; IF: 8.7)
5. Taslim Arefin Khan, Tarik Reza Toha, **Saiful Islam Salim**, Md Toki Tahmid and A. B. M. Alim Al Islam, “*Escalating Post-Disaster Rescue Missions Through Ad-hoc Victim Localization Exploiting Wi-Fi Network*”, Heliyon, Elsevier, 2022. (Q1 in Scimago; IF: 4.0)

6. **Saiful Islam Salim***, Najla Abdulrahman Al-Nabhan*, Masfiquir Rahaman, Nafisa Islam, Tarik Reza Toha, Jannatun Noor, Aaiyeesha Mostak, Mainul Hossain, Md. Masum Mushfiq, Adnan Quaium, and A. B. M. Alim Al Islam, “*Human-Survey Interaction (HSI): A Study on Integrity of Human Data Collectors in A Mass-Scale Hajj Pilgrimage Survey*”, IEEE ACCESS, IEEE, 2021. (Q1 in Scimago; IF: 3.9)
7. Jannatun Noor, **Saiful Islam Salim**, and A. B. M. Alim Al Islam, “*Strategizing Secured Image Storing And Efficient Image Retrieval through a New Cloud Framework*”, Journal of Network and Computer Applications (JNCA), Elsevier, 2021. (Q1 in Scimago; IF: 8.7)
8. Kazi Hasan Zubaer, Quazi Mishkatul Alam, Tarik Reza Toha, **Saiful Islam Salim**, and A. B. M. Alim Al Islam, “*Towards Simulating Non-lane based Heterogeneous Road Traffic of less Developed Countries using Authoritative Polygonal GIS Map*”, Simulation Modelling Practice and Theory (SIMPAT), Elsevier, 2020. (Q1 in Scimago; IF: 4.2)

POSTER

1. **Saiful Islam Salim**, Uday Kamal, Adnan Quaium, Mainul Hossain, Masfiquir Rahaman, Nazmul Hasan Sakib, and A. B. M. Alim Al Islam, “*An Economical Long-Range Sensing System for Real-Time Monitoring of Rail Tracks*”, ACM NSysS’21. Dhaka, Bangladesh, 2021. [1st **Runner-up**]

MENTORING EXPERIENCE

- Former Student Collaborators: Alexander E. Cooper (UofA), Suryashree Ray (UofA), Logan Michael Sandlin (UofA), Genesis Elizabeth Benedith (UofA), Xin Li (UofA), Mainul Hossain Yaad (BUET, Currently @George Mason University), Masfiquir Rahaman (BUET, Currently @UC San Diego), Nazmul Hasan Sakib (BUET), Kazi Ragib Ishraq Sanim (BUET, Currently @University of Cambridge)

REFERENCE

- **Dr. Mihai Surdeanu, Ph.D.**
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- **Dr. Marvin J. Slepian, M.D., J.D.**
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